METRIC

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PERFORMANCE SPECIFICATION

HYDRAULIC FLUID, FIRE RESISTANT, LOW TEMPERATURE, SYNTHETIC HYDROCARBON BASE, AIRCRAFT AND MISSILE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the characteristics and provides the requirements for a synthetic hydrocarbon base hydraulic fluid for use in the -54 °C to +200 °C temperature range in aircraft and missile hydraulic systems. This hydraulic fluid is identified by NATO Code No. H-538 (see 6.8).

Comments, suggestions, or questions on this document should be addressed to AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB, OH 45433 or e-mailed to AFPET.PTPS@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9150

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-791 - Testing Method of Lubricants, Liquid Fuels, and Related Products

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-5606* - Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance, NATO Code Number H-515

MIL-PRF-6083 - Hydraulic Fluid, Petroleum Base, for Preservation and Operation, NATO Code Number C-635

MIL-PRF-46170 - Hydraulic Fluid, Rust Inhibited, Fire Resistant, Synthetic Hydrocarbon Base, NATO Code Number H-544

MIL-PRF-83282 - Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537

* Indicates document is inactive for new design

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEFENSE FORMS MANAGEMENT PROGRAM

DD Form 1718 - Certification of Qualified Products

(Copies of this document are available online at https://www.esd.whs.mil/DD/.)

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DEFENSE STANDARDIZATION PROGRAM OFFICE

SD-6 - Provisions Governing Qualification

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D92 - Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester

ASTM D97 - Standard Test Method for Pour Point of Petroleum Products

ASTM D130 - Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test

ASTM D445 - Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)

ASTM D664 - Standard Test Method for Acid Number of Petroleum Products by Potentiometric Titration

ASTM D892 - Standard Test Method for Foaming Characteristics of Lubricating Oils

ASTM D972 - Standard Test Method for Evaporation Loss of Lubricating Greases and Oils

ASTM D1500 - Standard Test Method for ASTM Color of Petroleum Products (ASTM Color Scale)

ASTM D1298 - Standard Test Method for Density, Relative Density, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method

ASTM D2532 - Standard Test Method for Viscosity and Viscosity Change After Standing at Low Temperature of Aircraft Turbine Lubricants

ASTM D4052 - Standard Test Method for Density, Relative Density, and API Gravity of Liquids by Digital Density Meter

ASTM D4172 - Standard Test Method for Wear Preventive Characteristics of Lubricating Fluid (Four-Ball Method)

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ASTM D4289 - Standard Test Method for Elastomer Compatibility of Lubricating Greases and Fluids

ASTM D4636 - Standard Test Method for Corrosiveness and Oxidation Stability of Hydraulic Oils, Aircraft Turbine Engine Lubricants, and Other Highly Refined Oils

ASTM D5185 - Standard Test Method for Multielement Determination of Used and Unused Lubricating Oils and Base Oils by Inductively Coupled Plasma Atomic Emission Spectrometry (ICP-AES)

ASTM D5306 - Standard Test Method for Linear Flame Propagation rate of Lubricating Oils and Hydraulic Fluids

ASTM D5949 - Standard Test Method for Pour Point of Petroleum Products (Automatic Pressure Pulsing Method)

ASTM D6304 - Standard Test Method for Determination of Water in Petroleum Products, Lubricating Oils, and Additives by Coulometric Karl Fischer Titration

ASTM D7647 - Standard Test Method for Automatic Particle Counting of Lubricating and Hydraulic Fluids Using Dilution Techniques to Eliminate the Contribution of Water and Interfering Soft Particles by Light Extinction

(Copies of these documents are available from https://www.astm.org/.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 2230 - Rubber Products – Guidelines for Storage

ISO 4406 - Hydraulic fluid power — Fluids — Method for coding the level of contamination by solid particles

ISO 11171 - Hydraulic Fluid Power – Calibration of Automatic Particle Counters for Liquids

ISO 11218 - Aerospace – Cleanliness classification for hydraulic fluids

ISO 11500 - Hydraulic Fluid Power – Determination of the Particulate Contamination Level of a Liquid Sample by Automatic Particle Counting Using the Light-extinction Principle

(Copies of these documents are available from https://www.iso.org/.)

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AEROSPACE INDUSTRIES ASSOCIATION / NATIONAL AEROSPACE STANDARDS (AIA/NAS)

AIA/NAS1638* - Cleanliness Requirements of Parts Used In Hydraulic Systems

* Indicates document is inactive for new design after 30 May 2001.

(Copies of this document are available from https://www.aia-aerospace.org/.)

SAE INTERNATIONAL

SAE AS4059 - Aerospace Fluid Power – Contamination Classification for Hydraulic Fluid

SAE AMS3217/2 - Test Slabs, Acrylonitrile Butadiene (NBR-L) Low Acrylonitrile, 65-75

SAE AMS3400 - Fluid, Reference for Testing Polyalphaolefin (PAO) Resistant Material

(Copies of these documents are available from https://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The hydraulic fluid furnished under this specification shall be product that is authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3). Changes shall not be permitted in the formulation of an approved product unless specific, written approval of the qualifying activity is obtained.

3.2 Materials. The materials used in formulating this hydraulic fluid shall consist of synthetic hydrocarbon base and shall contain additives as specified in 3.2.1. A base oil mixture of 2 mm^2/sec and 4 mm^2/sec polyalphaolefin is suggested but is not mandatory. The hydraulic fluid shall contain no admixture of resins, soaps, gums, fatty oils, or oxidized hydrocarbons. A red dye (see 3.2.1.4) shall be used for coloring.

3.2.1 Additives. There shall be no restriction on the types of materials used as additives in the fluid except for those listed and those imposed by technical requirements of this specification. No pour point depressant materials or viscosity index improvers shall be used.

3.2.1.1 Oxidation inhibitors. Oxidation inhibitors of the phenolic type shall not exceed a concentration of 2.0 percent by weight.

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3.2.1.2 Anti-wear agent. Anti-wear agents shall be blended in sufficient quantity, not to exceed 3.0 weight percent, to permit the finished oil to meet four-ball requirements specified in Table I. Tricresyl phosphate (TCP) shall not be used.

3.2.1.3 Blending fluid. Blending fluids, such as diesters, shall be blended in a concentration not to exceed 35 percent by weight of the finished fluid but in sufficient quantity to meet the rubber swell, as specified in Table I, and viscosity requirement at -54 °C as specified in Table I.

3.2.1.4 Red dye. The fluid shall contain red dye in concentration not greater than one (1) part of dye per 10,000 parts of oil by weight.

3.3 Properties of finished fluid. The properties of the finished fluid shall be as specified in Table I and 4.4.

3.3.1 Color. There shall be no readily discernible difference in the color of the finished fluid compared to the standard color when tested as specified in 4.4.5.

3.3.2 Compatibility. The hydraulic fluid shall be compatible in all concentrations with each of the fluids approved under this specification, and miscible with hydraulic fluids conforming to MIL-PRF-5606, MIL-PRF-6083, MIL-PRF-46170, MIL-PRF-83282, and MIL-PRF-87257 fluids in all proportions from -54 °C to 200 °C. No formation of resinous gums, sludge or insoluble material shall occur (see 4.4.2).

3.3.3 Low temperature stability. The hydraulic fluid shall show no evidence of gelling, clouding, crystallization, solidification, or separation of ingredients. The presence of a slight haze that dissipates shall not be cause for failure.

3.3.4 Relative density. The relative density shall be determined and reported at qualification but shall not be limited. The relative density of conformance samples shall fall within ± 0.008 from that obtained at qualification.

3.3.5 Storage stability. The fully blended product shall show no separation of ingredients or evidence of crystallization. The fully blended product shall be clear and transparent when examined visually and shall conform to the requirements as specified in 3.3 after 12 months of storage.

3.4 Workmanship. The workmanship shall be in accordance with (IAW) high-grade, commercial practices covering this type of material. The finished fluid shall be homogeneous and free from suspended matter, grit, or other adulteration.

3.5 Toxic products and formulations. The hydraulic fluid shall have no adverse effect on the health of personnel when used for its intended purpose.

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TABLE I. Properties of finished fluid.

Property	Qualification[1]	Conformance[2]	Limits / Requirement Paragraph	Test Method		
				FED-STD-791	ASTM (ISO)	Test Paragraph
Acid number, mg KOH/g (max)	X	X	0.20		D664	
Barium content, mg/kg (max)	X	X	10		D5185	
Color	X	X	section 3.2.1.4 [3]		D1500	4.4.4
Compatibility	X		section 3.3.2 [4]			4.4.1
Corrosiveness and oxidation stability (168 hours @ 135 °C ± 1 °C)	X				D4636 [5]	
Change in acid number, mg KOH/g (max)			0.20			
Percent change in viscosity @ 40 °C			± 10			
Metal specimen weight change, mg/cm² [6]						
1010 Steel			± 0.2			
Aluminum			± 0.2			
Magnesium			± 0.2			
Cadmium plated steel [7]			± 0.2			
Copper [8]			± 0.6			
Separation of insoluble materials or gumming of the fluid			None			
Evaporation loss (6.5 hours @ 135 °C), weight percent (max)	X	X	20		D972	
High temperature-high pressure spray ignition	X		[9]	6052		
Flame propagation cm/sec (max)	X		0.5		D5306	
Flash point, °C (min)	X	X	160		D92	
Fire point, °C (min)	X	X	170		D92	
Foaming characteristics @ 24 °C	X	X			D892	
Foaming tendency, mL (volume at end of five-minute blowing period)			65			
Foam stability, mL (volume at end of ten-minute settling period)			0 [10]			

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TABLE I. Properties of finished fluid - Continued.

Property	Qualification[1]	Conformance[2]	Limits / Requirement Paragraph	Test Method		
				FED-STD-791	ASTM (ISO)	Test Paragraph
Four-ball, use test condition A, scar diameter, mm (max)					D4172	
1 kg load	X		0.21			
10 kg load	X		0.30			
40 kg load	X		0.65			
High temperature stability	X					4.4.2
Percent change in viscosity @ 40 °C (max)			± 5			
Change in acid number, mg KOH/g (max)			0.1			
Formation of precipitate or insoluble material			None			
Isothermal secant bulk modulus @ 40 °C and 27.6 MPa (4000 psig), MPa (psi) (min)	X		1379 (200,000)			4.4.7
Low temperature stability, 72 hours @ -54 °C ±1 °C	X	X	section 3.3.3 [11]	3458		
Particulate contamination						
Particle count, AS4059 Contamination Class [12] (max)		X	5 [13]	3012 [14][15]	(11500)	4.4.3.1
Particle counting, individual channel counts & ISO Code [12]		X	16/14/11[17]	3012	D7647[16] (11500)	4.4.3.1
Pour point, °C (max)	X	X	-60		D97, D5949[16]	
Relative Density @ 15.6°C/15.6°C	X	X	section 3.3.4 [1] [2]		D4052[16], D1298	
Rubber swell, standard synthetic rubber, NBR-L [18], percent change in volume	X	X	19.0-30.0		D4289[19]	4.4.5
Storage stability (24 °C ±3 °C for 12 months)	X		section 3.3.5[20]			4.4.6
Viscosity stability @ -54 °C, mm²/sec					D2532	
3 hours (max)	X	X	2500			
72 hours (max)	X		2500			
Viscosity, mm²/sec					D445	
@ 40 °C (min)	X	X	6.7			
@ 100 °C (min)	X	X	2.0			
@ -40 °C (max)	X	X	550			

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TABLE I. Properties of finished fluid - Continued.

Property	Qualification[1]	Conformance[2]	Limits / Requirement Paragraph	Test Method		
				FED-STD-791	ASTM (ISO)	Test Paragraph
Water, mg/kg (max)	X	X	100		D6304[21]	
Workmanship	X	X	section 3.4[22]			
NOTES:						
1.	Qualification inspection: Shall be determined but shall not be limiting.					
2.	Conformance inspection: Shall be ±0.008 of the value determined during qualification inspection.					
3.	There shall be no readily discernible difference between the color of the finished fluid and the standard color.					
4.	Compatible in all concentrations with each of the fluids approved under this specification. Shall be miscible with MIL-PRF-5606, MIL-PRF-6083, MIL-PRF-46170, and MIL-PRF-83282 fluids in all proportions from -54 °C to +200 °C. No formation of resinous gums, sludge, or insoluble material shall occur.					
5.	Use Alternative Procedure 2.					
6.	There shall be no pitting, etching, or visible corrosion on the surface of the metals when viewed under magnification of 20 diameters.					
7.	A slight discoloration shall be permitted.					
8.	Any corrosion (discoloration) produced on the surface shall not exceed that of a No. 3 rating of the ASTM D130 copper strip corrosion standard.					
9.	The flame shall not continue to burn when the source of ignition is removed.					
10.	A ring of small bubbles around the edge of the graduated cylinder shall be considered complete collapse.					
11.	No evidence of gelling, clouding, crystallization, solidification, or separation of ingredients.					
12.	Particulate contamination shall be measured as either particle count, AS4059 Contamination Class or particle counting, individual channel counts & ISO Code.					
13.	The container shall be thoroughly shaken immediately prior to withdrawing the sample. A determination shall be made on 100-mL portion of the sample.					
14.	See Table II for particle count per size range.					
15.	Hexane shall be used instead of ozone depleting solvents.					
16.	Referee Method.					
17.	Results shall be reported for particle size, channel count, and ISO code in same format and criteria as laid out in Table III.					
18.	The test shall be run at 70 °C ± 2.5 °C for 168 hrs ± 0.5 hrs.					
19.	The fully blended product shall be clear and transparent, with no separation of ingredients or evidence of crystallization, and shall conform to all requirements of this specification.					
20.	The fully blended product shall be clear and transparent, with no separation of ingredients or evidence of crystallization, and shall conform to all requirements of this specification.					
21.	Use Procedure A (Direct Injection).					
22.	Homogeneous and free from suspended matter, grit, or other adulteration.					

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4. VERIFICATION

4.1 Classification of inspections. The examination and testing of the hydraulic fluid are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall consist of testing to all the requirements specified in sections 3.3 thru 3.4 and Table I. When required by the qualifying activity, additional evaluations may be required on candidate formulations.

4.2.1 Qualification process. At the initiation of the qualification process, prospective suppliers will forward a written request for such action to the qualifying activity, Air Force Petroleum Office, AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB, OH 45433 or via email. This request will contain general information on the proposed candidate material. The qualifying activity will reply in writing outlining the required information to be submitted, the governing regulations for qualified products, and the cost of testing. The information that will be submitted includes:

a. Letter stating the company will adhere to the appropriate provisions of the SD-6 on company letterhead.

b. Complete formulation of the candidate product, including chemical composition in weight percent, the manufacturer, the commercial trade name and CAS number, the purities, and use of each component.

c. Safety Data Sheet (SDS) for each component and final product. The SDS will not be older than 5 years.

d. Certified laboratory test data from the manufacturer, a commercial laboratory, or combination thereof showing quantitative results of all qualification tests required by this specification as identified in Table I.

e. Verification that the System for Award Management (SAM) (CAGE Codes for the corporate office and each plant) is current. This can be viewed at https://sam.gov/.

f. Identification of the manufacturing site of the specific batch of test sample to be submitted.

g. A completed DD Form 1718.

4.2.2 Qualification inspection sample. A qualification sample shall consist of a one (1) gallon container of hydraulic fluid. The qualifying activity will request manufacturer and third-party data prior to authorizing submission of the qualification sample.

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4.2.3 Retention of qualification. To retain qualification of products approved for listing on the Qualified Products List (QPL), the manufacturer will be requested to verify by certification (DD Form 1718) to the qualifying activity that its product(s) comply with the requirements of this specification. The manufacturer shall also provide a recent Certificate of Analysis (COA), to include all conformance inspection requirements (see 4.3), a current SDS, and formulation. Unless otherwise specified by the qualifying activity, the time of periodic verification by certification will be in 2 1/2 year intervals from the date of original qualification and will be initiated by the manufacturer. The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine that the product continues to meet any or all of the specification requirements.

4.2.4 Requalification. Requalification shall be required every five (5) years or when any change is made in the source of manufacture, purity, or composition of the fluid's base stock or additives. A minor change in the fluid formulation such as a new supplier of stock material might not need requalification testing, but only after notification to, and approval by the qualifying activity (see 6.3). The documentation required for requalification are the same as requirements stated in section 4.2.1 and shall include product formulation, safety data sheet (SDS) for each component and final product, a COA showing compliance to all of the qualification requirements specified in Table I and a one (1) gallon requalification sample from a recent production run within the previous 60 days. The SDS will not be older than five (5) years. The qualifying activity may, at its discretion, waive complete requalification or may require only partial requalification testing to determine the significance and acceptability of the proposed formulation change.

4.3 Conformance inspection. The conformance inspection (Government Lot Acceptance Testing) shall be conducted on samples from randomly selected containers of each packaged lot (see 6.4.2) and consists of tests to verify conformance to the specific conformance requirements in Table 1 and sections 3.3 thru 3.4. Samples shall be labeled completely with information identifying the purposes of the sample, name of product, specification number, lot and batch number, date of manufacture, date of sampling, and contract number.

4.4 Method of inspection. Tests shall be performed in accordance with the methods specified in Table I and section 4. Physical and chemical values, specified in section 3, apply to the arithmetic average of the determinations made on the samples for those values which fall within any stated repeatability or reproducibility limits of the applicable test method.

4.4.1 Compatibility. Samples of candidate fluid in amounts of 20 mL, 100 mL, and 180 mL shall be mixed with samples from each of the fluids previously approved under this specification. Total volume of each mixture shall be 200 mL. Mixtures shall be prepared in 250 mL stoppered flasks. The flasks shall be thoroughly agitated and then stored in an oven at 200 °C for 2 hours. At the end of this time, none of the mixtures shall show any sign of sediment, haziness, or crystallization. The samples shall then be stored at -54°C for a period of 2 hours. A slight haze that dissipated will be permitted in the samples at this time. Compatibility tests described herein shall also be conducted with representative fluids qualified to MIL-PRF-5606, MIL-PRF-6083, MIL-PRF-46170, and MIL-PRF-83282.

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4.4.2 High-temperature stability.

4.4.2.1 Apparatus. Thermal-stability characteristics of the hydraulic fluid shall be determined using the following apparatus: a 50 mL round-bottom flask or test tube (25 mm x 50 cm) with a 24/40 taper (Ace Glass Inc., Vineland, NJ 08360 or equivalent); a 75° angle "Y" side arm adapter with standard 24/40 taper (Ace Glass 5040-10 or equivalent); borosilicate glass gas inlet tube (Koehler K351-0-3 or equivalent); PTFE thermocouple adapter, 24/40, ¼" probe (ChemGlass CG-1048-A-05 or equivalent); aluminum foil collar (if necessary); 6-mm Outside Diameter (OD) stainless steel tubing, type 304, 2.5 cm in length; a nitrogen gas source (dry, lamp grade) with regulator; a calibrated mass flow meter; a calibrated digital thermometer; and a high-temperature bath capable of maintaining a temperature of 200 °C ± 2 °C.

4.4.2.2 Procedure. The following procedures shall be followed for determining high-temperature stability:

a. Place a 2.5-cm length of 6mm OD stainless steel tubing, type 304, into a 50 mL round-bottom flask or 25 mm x 50 cm test tube.

b. Pour 25 mL of candidate fluid into the flask or test tube.

c. Fit the flask or test tube with a 75° angle adapter.

d. Slide the PTFE thermocouple adapter onto the gas inlet tube and insert the gas inlet tube into the flask or test tube assembly, keeping the tip of the gas inlet tube above the surface of the fluid. Adjust the height of the gas inlet tube so that the tip will be approximately 1.0 cm above the surface level of the heat-expanded fluid.

e. Place the flask or test tube assembly into a high-temperature bath preheated to 200 °C ± 2 °C, using a calibrated digital thermometer to monitor temperature. If necessary, use a collar made of aluminum foil on the top of the flask or test tube to ensure the apparatus is stable and centered in the bath.

f. Use a calibrated mass flow meter to ensure the dry nitrogen (lamp grade) source is at a flow rate of 1-liter-per hour, then connect it to the gas inlet tube on the apparatus.

g. Heat the fluid for 100 hours at a temperature of 200 °C ± 2 °C while maintaining a 1-liter-per hour flow of dry nitrogen (lamp grade) on the surface of the fluid via the gas inlet tube positioned within 1.0 cm above the level of the heat-expanded fluid.

h. Remove the apparatus from the high-temperature bath and allow the fluid to cool to room temperature.

4.4.2.3 Evaluation. Upon completion of the test, the fluid shall meet the requirements of Table I.

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4.4.3 Particulate contamination.

4.4.3.1 Particle count. Particle count shall be determined as follows:

a. An automatic particle counter employing either the light extinction principle or optical imaging and meeting the channel, particle size range, and calibration requirements specified in Table II or Table III shall be used (use the operating instructions for the applicable instrument). The instrument shall meet the calibration requirements of ISO 11171.

b. Mix the sample thoroughly immediately prior to withdrawing the sample.

TABLE II. Particle Count, SAE AS4059/ISO 11218 Contamination Level 5.

Particle Size Ranges, micrometers (µm)[1] AIA/NAS 1638[2] (ISO 11171)		Maximum number of particles per 100 mL
SAE AS4059	SAE AS4059	DIFFERENTIAL
5-15	6-14(c)	8000
16-25	15-21(c)	1425
26-50	22-38(c)	253
51-100	39-70(c)	45
>100	>70(c)	8
SAE AS4059	SAE AS4059	CUMULATIVE
>5	>6(c)	9731
>15	>14(c)	1731
>25	>21(c)	306
>50	>38(c)	53
>100	>70(c)	8

NOTES:

1. The symbol (c) is used to designate the automatic particle counter was calibrated to ISO 11171.
2. AIA/NAS 1638 - Cleanliness Requirements of Parts Used in Hydraulic Systems is obsolete and replaced with SAE AS4059. Referenced for transition purposes only.

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TABLE III. Particle Count, ISO 4406 (SAE AS4059/ISO 11218 Contamination Level 5 Basis).

Particle Size Ranges, micrometers (µm)[1]		
Size	Counts per mL	ISO Code[3]
≥ 4 µm(c)	[2]	Max 16
≥ 6 µm(c)	[2]	Max 14
≥ 14 µm(c)	[2]	Max 11
≥ 21 µm(c)	[2]	[2]
≥ 38 µm(c)	[2]	[2]
≥ 50 µm(c)	[2]	[2]
≥ 70 µm(c)	[2]	[2]

NOTES:	
1.	The symbol (c) is used to designate the automatic particle counter was calibrated to ISO 11171.
2.	Report.
3.	Per Table 1 “Allocation of scale numbers” of ISO 4406.

4.4.4 Color of the finished product. The color of the hydraulic fluid shall be compared with a standard sample prepared by adding one (1) part dye to 10,000 parts of an oil not darker than ASTM D1500 color number 1.0.

4.4.5 Rubber swell. Rubber swell shall be performed IAW ASTM D4289 with the following modifications:

a. Use the standard synthetic rubber, NBR-L meeting the latest revision of SAE AMS3217/2. Store in the dark, in an inert atmosphere, below 65% relative humidity, away from direct heat sources, and between 15 °C to 25 °C (59 °F to 77 °F) IAW ISO 2230.

b. Check elastomer for conformance to SAE AMS3217/2, latest revision, at least every 6 months from the date of manufacture using SAE AMS3400 as the reference fluid.

c. Age the specimens by storing the container in a convection oven at 70 °C ± 2.5 °C for 168 hrs ± 0.5 hrs.

4.4.6 Storage stability.

4.4.6.1 Apparatus. Storage stability characteristics of the hydraulic fluid shall be determined using the following test apparatus: A storage chamber that is able to maintain a constant-temperature of 24 °C ± 3 °C (75 °F ± 5 °F) and is free of vibration (such as caused by heavy traffic). A 3.8 L (1 gal) clear-glass wide-mouth jar with cap.

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4.4.6.2 Procedure. The following procedures shall be followed for determining storage stability:

a. Clean the jar and cap by scrubbing with scouring powder and rinsing with tap water. Complete the rinsing with distilled water and air dry the jar and cap (using an oven at 104 °C (220 °F) if desired).

b. Line the cap with aluminum foil.

c. Pour approximately 3.8 L of the sample into the jar, and replace and tighten the cap.

d. Wrap aluminum foil around the jar to exclude all light from the sample.

e. Identify the contents of the glass jar (including type of sample, date stored, and any other useful information).

f. Store the sample for 12 months (one year) in the storage chamber at 24 °C ± 3 °C (75 °F ± 5 °F).

g. At the end of the 12-month period, remove the jar from the storage chamber and take off the aluminum wrapper, being careful not to shake or agitate the contents.

h. Visually examine the sample, and note the presence of cloudiness, sediment, suspended matter, discoloration, or other changes in homogeneity.

4.4.6.3 Evaluation. Upon completion of the test, the fluid shall meet the requirements of Table I.

4.4.7 Bulk modulus.

4.4.7.1 Apparatus. The bulk modulus of the hydraulic fluid shall be determined using a calibrated, precision capillary pycnometer of the type shown in Figure 1 (this is a modified 21 T 50 Jerguson pressure gauge). A sample pressure vessel and auxiliary equipment for this determination are shown in Figures 2 and 3.

4.4.7.2 Procedure. The pycnometer volume to capillary diameter ratio shall be chosen to provide a precision of measurement for liquid density of ±2 parts in 10,000. The pycnometer shall be charged with candidate fluid to the top of the capillary at 40°C and atmospheric pressure. Insert the pycnometer in the high-pressure pycnometer housing such that the capillary tube is visible through the window of the pressure vessel and parallel to the outside vertical surfaces of the pressure vessel. Assemble the pressure vessel containing the pycnometer, place it in a 40°C constant-temperature bath, allow equilibrium to be reached, and take a volume reading at atmospheric pressure. Since the precision of the unit depends upon visual readings, avoid errors due to parallax and distortion in the pressure vessel window and the walls of the constant-temperature bath. Increase nitrogen pressure to a new level and, after a 60-minute soak, take a third reading.

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For any pressure range, the secant bulk modulus is defined as follows:

Bulk modulus = $[(V \times DP) \div (DV + DV_g)]$

Where:

V is the original volume of the fluid,
DV is the observed volume change due to DP increase in pressure,
DP is the pressure change between the two measurements in kPa, and
DV_g is the correction factor, which considers the bulk modulus of glass in determining the true volume of the pycnometers at pressures above atmospheric. The bulk modulus of Pyrex glass is 3.28×10^7 kPa. $DV_g = [(V \times DP) \div 3.28 \times 10^7]$.

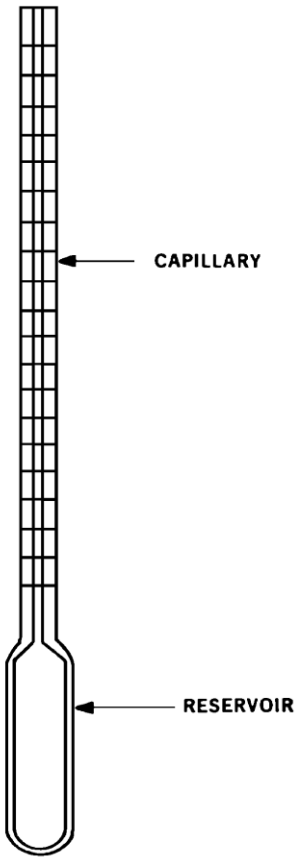


FIGURE 1. Precision capillary pycnometer

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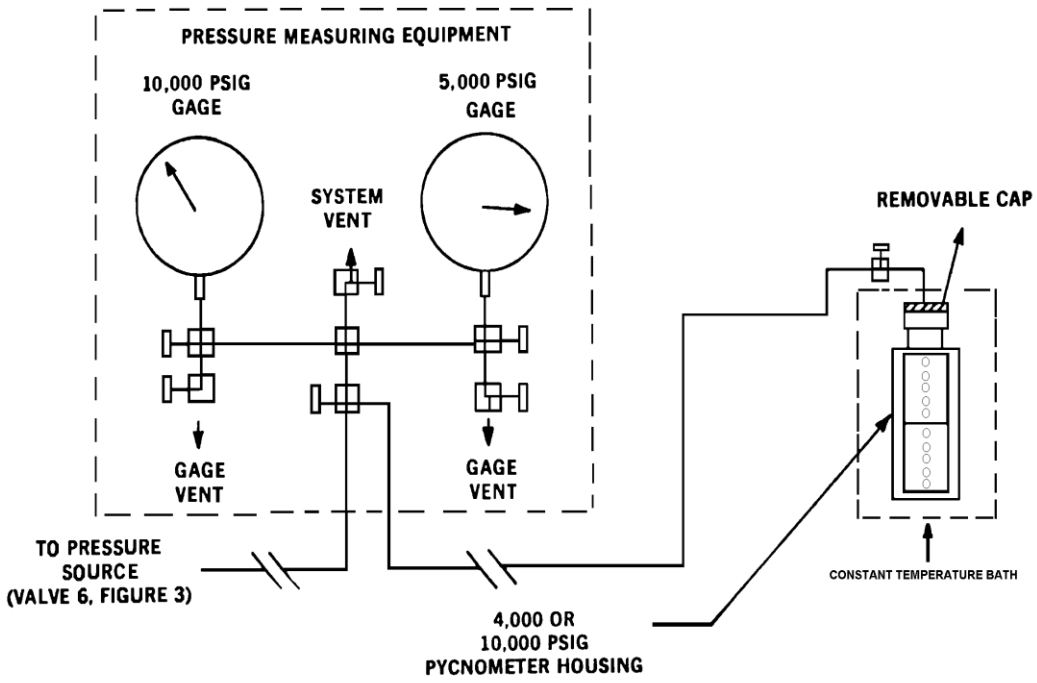


FIGURE 2. Diagram of bulk modulus equipment.

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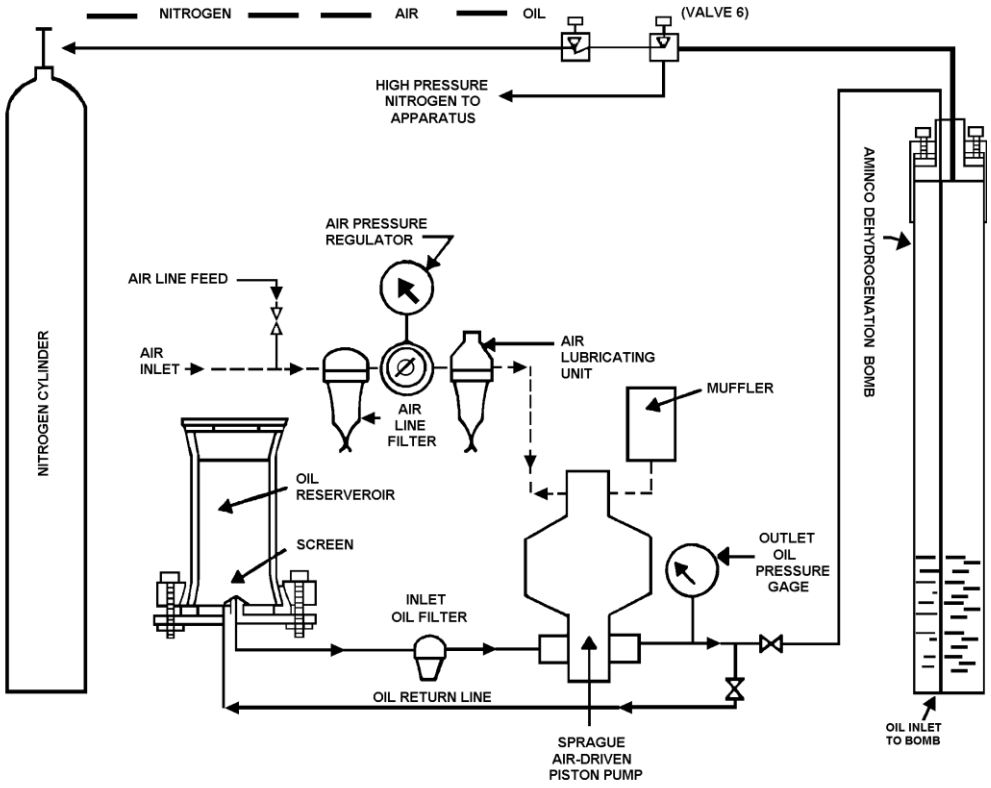


FIGURE 3. Auxiliary equipment.

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4.4.7.3 Evaluation. Upon completion of the test, the fluid shall meet the requirements of Table I.

4.4.8 Examination of filled containers. A random sample of filled containers from each packaged lot (see 6.4.2), shall be examined with regard to fill closure, sealing, and leakage.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hydraulic fluid covered by this specification is intended for use from -54 °C to +200 °C in automatic pilots, shock absorbers, brakes, flap-control mechanisms, missile hydraulic servo-controlled systems, and other hydraulic systems using synthetic sealing material.

6.1.1 Storage conditions. Prior to use in the intended equipment, the product may be stored under conditions of covered or uncovered storage in geographic areas ranging in temperatures from -57 °C to +49 °C (-70 °F to +120 °F).

6.1.2 Interchangeability. This fluid is completely compatible with MIL-PRF-5606, MIL-PRF-6083, MIL-PRF-46170, and MIL-PRF-83282 hydraulic fluids. It may be interchangeable with these fluids for some applications. The selection of the fluids to be used depends on the requirements of the operational system.

6.1.3 Cleaning agents. Chlorinated solvents should not be used for testing or cleaning hydraulic components. Residual solvent contaminates the hydraulic fluid and may lead to corrosion. The recommended solvents are MIL-PRF-680, Type II; A-A-59601, Type II; or other petroleum distillate type solvents.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size of containers.

c. Quantity.

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, contract awards will be made only for such products that are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 87257 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB OH 45433; e-mail: AFPET.PTPS@us.af.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Qualification information. It is understood that the material furnished under this specification subsequent to final approval will be of the same composition and will be equal to products on which approval was originally granted. In the event that fluid furnished under contract is found to deviate from the composition of the approval product, or that the product fails to perform satisfactorily, approval of such products may be subjected to immediate withdrawal from the QPL at the discretion of the approving activity.

6.3.2 Proceed authorization notification. Upon successful review of the submitted information, the Air Force Petroleum Office will issue authorization to ship the qualification sample(s) for testing.

6.3.3 Qualification samples. Upon receiving authorization from the Air Force Petroleum Office, qualification sample(s) may be shipped to the qualification activity, as detailed in the authorization letter. Sample(s) should be accompanied by an SDS and a copy of the authorization letter. Each sample should be plainly identified by securely attached durable tags or labels marked with the following information:

a. QUALIFICATION SAMPLE

b. Hydraulic Fluid, Fire Resistant; Low Temperature, Synthetic Hydrocarbon Base, Aircraft and Missile

c. MIL-PRF-87257

d. Name of Manufacturer

e. Product Code Number

f. Batch or Lot Number

g. Date of Manufacture

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6.4 Definitions.

6.4.1 Bulk lot. A bulk lot (batch) is an indefinite quantity of a homogeneous material mixture offered for acceptance in a single, isolated container or manufactured in a single-plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material (see 6.3.3).

6.4.2 Packaged lot. A packaged lot is an indefinite number of unit containers of identical size and type offered for acceptance and filled with a homogeneous material mixture from one isolated container; or filled with a material mixture manufactured in a single-plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

6.5 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings IAW MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, Volume 1, *Shelf-Life Management Program: Program Administration*; and Volume 2, *Shelf-Life Management Program: Materiel Quality Control Storage Standards*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.6 Subject term (key word) listing.

Additive

Anti-wear

Contamination

Flash point

Hydraulic system

Oxidation stability

Polyalphaolefin

Pour point

Tricresyl Phosphate

Viscosity

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6.7 Safety and handling.

6.7.1 Toxicity. Questions pertinent to this effect should be referred by the contracting activity to the appropriate departmental medical service who will act as its advisor.

6.7.2 Handling precautions. Personnel handling the product should wear appropriate impervious clothing to prevent repeated or prolonged skin contact. Local appraisal is required for exact health and safety implications and to prescribe precise application of protective clothing. If skin or clothing becomes moistened with the product, personnel should promptly wash with soap or mild detergent and water. Respirators are not required unless there is an inhalation exposure to mists. Personnel should wear protective clothing when using the product and when cleaning up spills.

6.7.3 Resource recovery. The accumulated waste fluid should be disposed of through a waste oil recovery program unless prohibited by local law. Otherwise, the product should be disposed of IAW local law and regulations promulgated by the U.S. Environmental Protection Agency under Public Law 94-580, Resource Conservation and Recovery Act of 1976.

6.7.4 Depot-type operations. The used product which has been drained from the hydraulic systems should be combined with unused, but contaminated, fluid from partially full containers and then recycled.

6.7.5 Container disposal. Containers should be made as empty as possible using gravity draining, after which they are to be crushed and buried in a permitted sanitary landfill or incinerated with general refuse. No special decontamination procedures are required for empty containers or their lids.

6.8 Isothermal secant bulk modulus testing. Bulk modulus is a required test in support of product qualification under this specification. With the ASTM International withdrawal of ASTM D6793 *Standard Test Method for Determination of Isothermal Secant and Tangent Bulk Modulus*, an alternative test method has been incorporated in section 4.4.7 of this specification to support product qualification testing requirements. If at some future date, ASTM D6793 is reinstated by ASTM International, the test method will be re-incorporated into this specification.

6.9 International standardization agreement implementation. This specification implements:

a. Air Force Interoperability Council (AFIC) Air Standard Fuels, Lubricants, and Gases (FLG) 4023 *Hydraulic Fluid, Petroleum (H-515), Polyalphaolefin (H-537 and H-538), and Phosphate Ester Fluids (H-522, H-523, H-524)*.

b. AFIC Air Standard FLG 4024 *Interchangeability Chart of Standardised Aviation Fuels, Lubricants, and Associated Products*.

c. NATO STANAG/AFLP-1135 *Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations*.

d. NATO STANAG/AFLP-3748 *Hydraulic Fluid, Petroleum (H-515), Polyalphaolefin (H-537, H-538) and Phosphate Ester Fluids (H-522, H-523, H-524)*.

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When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.10 Conversion of metric units. Units of measure have been converted to the International System of Units (Metric) IAW IEEE/ASTM SI10 *American National Standard for Metric Practice*. If test results are obtained in units other than Metric or there is a requirement to report dual units, IEEE/ASTM SI10 or NIST SP 811 *Guide for the Use of the International System of Units (SI)* should be used to convert the units.

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL:

Custodians:
Army – AT
Navy – AS
Air Force – 68

Preparing activity:
Air Force – 68
(Project 9150-2023-010)

Review activities:
Army – AV, MD1
Navy – SH
Air Force – 20
DLA – GS, PS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information using the ASSIST Online database at https://assist.dla.mil.

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